REPORT

Routing report, rounds 1 and 2

> Historical (21 October 2026). Superseded for tables and page text by the final advisory.

Source: `merged/final/MERGED_REPORT.md`, produced by `tools/merge_results.py --round2` from the desktop export (22,731 rows) and the validated laptop export (18,244 rows, SHA-256 `ebc69a53...f5da753`, matches). Merged: 40,975 rows, 26 same-model comparison groups. Both exports passed `check_conformance.py --round2` (CONFORMANT). Everything below is scores and counts from code. No model output was read by a person or by the desktop operator. Failed cases count as zero.

Status: prototype routing, not production approval

These results select candidate routes for the prototype. They do not approve anything for production. The best laptop crop OCR (8B) still has about 1.23% character error and 96.5% numeric recall, below the project's existing acceptance targets. Readiness is decided by human-labelled institutional documents (the annotation pilot), not by these public benchmarks.

Routing choices

Task	Route	Measured basis
Crop OCR (prose crops)	Qwen3-VL 8B instruct, baseline	Character error 1.2% vs 2.0% for Qwen2.5-VL 7B (50 crops each, laptop). 8B matches the desktop 32B models (1.1% to 1.2%) and runs in 2.6 s median on the laptop
Charts, question answering	Qwen3-VL 8B instruct, **self-check off**, baseline or upscale	79.5% baseline (200 cases, laptop). Self-check drops it to 36.0%, a loss of 43 points (95% CI -51 to -35)
Charts, 7B fallback	Qwen2.5-VL 7B, **self-check on**	52.5% to 79.0%, a gain of 26.5 points (CI +18.5 to +34), and +17.8 points on the v2 confirmation set (1,322 questions across 722 charts; questions on the same chart are not independent)
Chart to table	8B with self-check or expert prompt	Score 0.42 to 0.93 (self-check), 60 cases. Baseline 8B is poor here
Tables	Any of the 8B variants, baseline is enough	TEDS 0.82 to 0.83, no variant differs significantly
Whole-page text masking (layoutmask)	**Experimental, not routed by default**	Helps 7B (+0.085) and the desktop 32B (+0.067); no gain on 8B (+0.010, CI includes 0). On the desktop 32B it lowers numeric recall: 0.909 to 0.720 on the 40 development pages, and 0.872 to 0.694 on the 80 confirmation pages (reported separately; different page sets)
Three-vote chart answers (vote3)	Use only with 7B or 32B	+0.045 on 7B (CI +0.030 to +0.061), +0.026 on 32B; no effect on 8B (-0.002, not significant)

Charts, question answering (relaxed accuracy, 200 development cases)

Model	Baseline	Expert	Self-check	Upscale
Qwen2.5-VL 7B, laptop	52.5%	47.5%	**79.0%**	47.5%
Qwen2.5-VL 7B, desktop	53.5%	44.5%	80.0%	49.0%
Qwen3-VL 8B instruct, laptop	**79.5%**	75.0%	36.0%	80.0%
Qwen3-VL 8B instruct, desktop	78.0%	76.0%	32.5%	79.5%
Qwen3-VL 32B, desktop (reference)	85.0%	84.0%	91.0%	89.5%

Round-2 confirmation (1,322 questions across 722 charts, laptop), change from baseline with 95% CI:

Model	Variant	Change	95% CI	Verdict
7B	vote3	+0.045	+0.030 to +0.061	better (confirmatory)
7B	self-check	+0.178	+0.150 to +0.207	better (supporting)
7B	upscale	-0.038	-0.055 to -0.021	worse
8B	vote3	-0.002	-0.011 to +0.008	not significant
8B	self-check	-0.427	-0.455 to -0.396	worse
8B	upscale	+0.015	+0.003 to +0.027	slightly better
32B, desktop	vote3	+0.026	+0.014 to +0.037	better (confirmatory)

Crop OCR (character error, lower is better, 50 crops)

Model	Laptop	Desktop
Qwen3-VL 8B instruct	**0.012**	0.012
Qwen2.5-VL 7B	0.020	0.018
32B-class models (desktop only)	n/a	0.011 to 0.012

Tables (TEDS, 60 cases) and chart data extraction

Model	Table TEDS laptop	Table TEDS desktop	Chart to table laptop (baseline / self-check)
Qwen3-VL 8B instruct	0.818	0.819	0.418 / 0.930
Qwen2.5-VL 7B	0.724	0.724	0.650 / 0.626

Page masking (word F1)

Model	Cases	Baseline	Layoutmask	Change [95% CI]
7B, laptop	80	0.757	0.843	+0.085 [+0.039, +0.135]
8B, laptop	80	0.828	0.838	+0.010 [-0.035, +0.050]
32B, desktop	80	0.814	0.881	+0.067 [+0.033, +0.101]

Do the two machines agree?

Qwen2.5-VL 7B has the same weights on both machines. Of its 13 comparisons, 11 reproduced; 2 disagreed (chart_qa expert, +3.0 points; chart_table upscale, -14.8 points, a 60-case task). Crop OCR and tables reproduced.

Caveats

- The 8B digest on the laptop (`70e58e37...b96eb`) differs from the desktop record. The merged report marks all 8B rows "different weights: not a reproduction test". The 8B comparison is a method comparison, not a reproduction. Even so, laptop and desktop 8B scores agree closely (for example crop error 0.0123 vs 0.0124).
- 7B and 8B results are not reproductions of the desktop 32B. The vote3 pick was made on 32B development data. On 7B and 8B it replicates the protocol with a different model, and does not count as a new selection.
- Many tasks have only 50 to 60 cases, so their intervals are wide. Holm-adjusted p-values are in the merged report. Only the round-2 confirmation set is large: 1,322 questions across 722 charts (questions cluster by chart, so the effective sample is smaller than 1,322) and 80 pages.

- Page masking stays experimental per the round-2 contract, and needs a numeric-recall check before use.
- Self-check's effect depends on the model and task. It helps 7B and the desktop 32B on chart answering, but hurts 8B. Route it per model and task, not globally.

Corrections, 21 Oct 2026 (Codex review): confirmation-set size stated as questions across charts; masking numeric-recall figures labelled by page set (40 development vs 80 confirmation); self-check wording changed to model- and task-dependent; prototype-not-production status added.

Public benchmarks only. No private documents, labels or model responses.

Source and frozen protocols